

(1)

$$Q^\pi(s, a) = E[Z^\pi(s, a)]$$

(2)

$$Z^\pi(s, a) = R(s, a) + \gamma Z^\pi(s', a')$$

$$\text{where } s' \sim p(\cdot | s, a) \quad a' \sim \pi(\cdot | s')$$

(3)

define the operator T^π acting on return distribution $Z(s, a)$

$$\Rightarrow T^\pi Z(s, a) = R(s, a) + \gamma Z(s', a')$$

$$\text{where } s' \sim p(\cdot | s, a), \quad a' \sim \pi(\cdot | s')$$

key properties of T^π

$$\begin{cases} T^\pi Z^\pi = Z^\pi \\ W_1(T^\pi Z_1, T^\pi Z_2) \leq \gamma W_1(Z_1, Z_2) \\ E[(T^\pi Z)(s, a)] = R_s^a + \gamma E_{s', a'}[E[Z(s', a')]] \end{cases}$$